University of Trento
Master's Degree in Computer Science

Data Visualization Technical Report

Data Visualization Lab
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Davide Donà	264467
Andrea Blushi	264468
Lorenzo Di Berardino	265984

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1 City Bike Dataset

For this project, our main dataset is the Citi Bike system dataset, a publicly available dataset tracking trips made through New York City's Citi Bike share network since 2013.

1.1 Dataset Structure

Released monthly as CSV files, these records log every completed ride, with high-volume months split across multiple files. Each row represents a single trip, offering detailed attributes that characterize the journey.

1.1.1 Original Structure

The original dataset includes the following columns:

- **Ride ID**: A unique identifier for each bike trip.
- **Rideable Type**: The type of bike used for the trip ("classic" or "electric").
- **Started At**: A timestamp indicating when the trip started.
- **Ended At**: A timestamp indicating when the trip ended.
- **Start Info**: Information about the station and location where the trip started, structured as:
  - **Station Name**: The name of the starting station.
  - **Station ID**: The identifier of the starting station.
  - **Latitude**: The latitude coordinate of the starting station.
  - **Longitude**: The longitude coordinate of the starting station.
- **End Info**: Information about the destination station, structured equally to Start Info.
- **User Type**: The type of user ("member" or "casual").

1.1.2 Extracted Features

Additional per-trip features were derived to support the analysis:

- **Trip Duration**: The duration of the trip in seconds, calculated as Ended At − Started At.
- **Trip Distance**: The straight-line distance between the start and end stations, computed using the Haversine formula and scaled by a circuity factor of 1.3 to approximate actual street routes.

Then, these individual trips were aggregated to derive:

- **Station Activity**: The total number of trips starting and ending at each station during a given time period, calculated by grouping trips by station.
- **Origin-Destination Flow**: The total number of trips made between each pair of stations within a given time period, calculated by grouping trips by station pair.

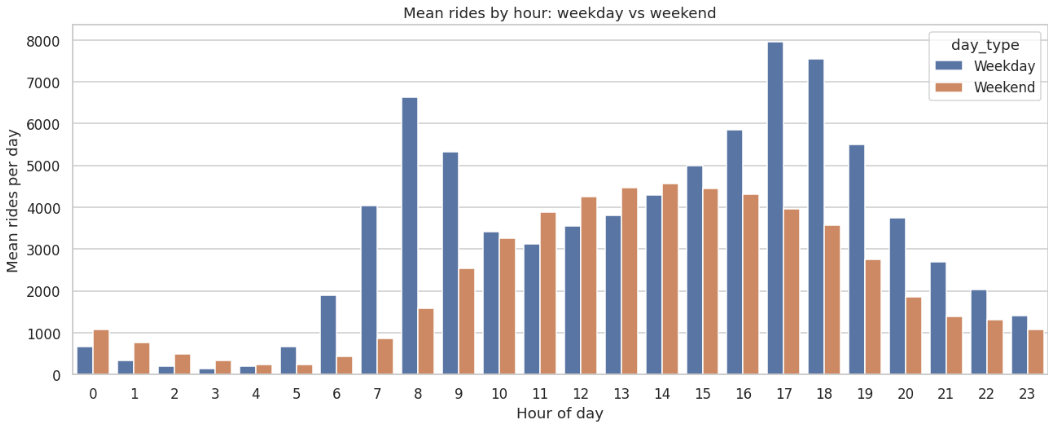

Figure 1: Distribution of trips by hour of the day and weekday/weekend.

1.2 Analysis and Insights

For transparency and reproducibility, the complete workflow is available in our Jupyter notebook. Using the February 2025 trip dataset, the following sections examine temporal, spatial, and behavioral usage patterns. Rather than offering a comprehensive study of the Citi Bike system, this prior analysis serves to justify our visualization design choices.

Temporal Analysis Figure 1 illustrates distinct variations in demand across different hours and days of the week. Weekday trips peak during standard commuting hours, whereas weekend usage shifts toward smooth patterns. Additionally, late-night hours (12:00 AM–3:00 AM) show slightly higher weekend volume, whereas weekday activity rises sharply starting at 6:00 AM. These temporal patterns provide a foundation for designing visualizations that highlight trip trends, enabling users to explore how demand fluctuates over time.

Spatial Analysis Figure 2 illustrates the spatial distribution of trip origins, covering the whole metropolitan area. As is possible to see, demand is heavily concentrated in Manhattan and high-density commercial corridors, where multiple grid cells exceed 5,000 trip starts within the observation window. A clear spatial decay occurs moving outward toward outer residential boroughs, where activity steadily tapers to lower tiers (0-500 trips per cell). In addition, distinct holes emerge within the heatmap that coincide with unserviced or non-drivable areas, such as Central Park. Those initial insights, together with the city coverage, provide a strong basis for asserting a spatial pattern in Citi Bike usage.

User/Bike Type Behaviour Analysis As shown in Figure 3, both casual riders and members prefer electric bikes. Because e-bikes allow users to cover greater distances faster, overall trip durations tend to be shorter. However, casual riders still take longer trips than members across both bike types, whereas members maintain consistently shorter trips regardless of vehicle choice. These contrasting behaviors highlight distinct usage patterns, useful to inform for example pricing strategies, marketing campaigns, and infrastructure planning.

Trip Duration and Distance Figures 4 and 5 provide additional insight into mobility behaviour. Most trips are relatively short, while longer rides are less fre-

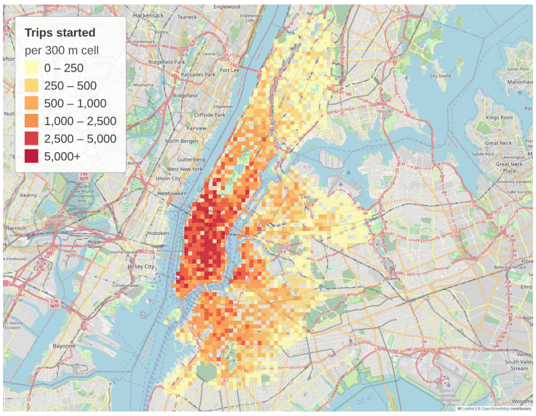

Figure 2: Distribution of trips by station of origin, aggregated into 300 m grid cells. Each color class maps to a specific trip volume tier, preventing visual clutter from individual stations.

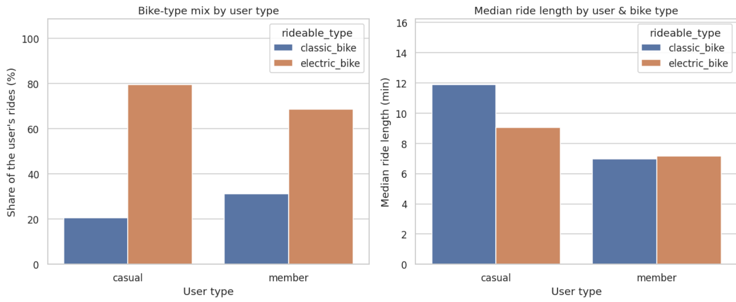

Figure 3: Distribution of trips by user/ride type based on rides and trip length.

quent: the median ride lasts 7.3 minutes (peaking at 4-5 minutes), while the median distance is 1.73 km (peaking near 1 km). It's important to note that durations begin at one minute, as Citi Bike removes trips shorter than 60 seconds at the source, since they are likely to be erroneous starts. To keep the charts readable without losing extreme values, long-tail outliers are grouped into the final bins (≥ 60 min and ≥ 10 km).

1.3 Data Quality

To prevent misleading visual representations and maintain data integrity, we identified several data quality issues in the raw Citi Bike dataset.

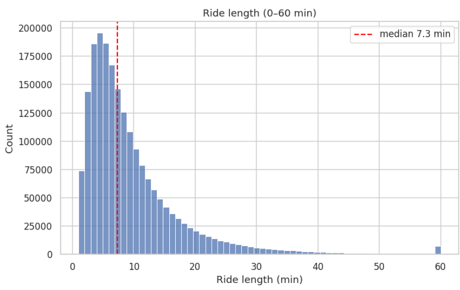

Figure 4: Distribution of trip durations.

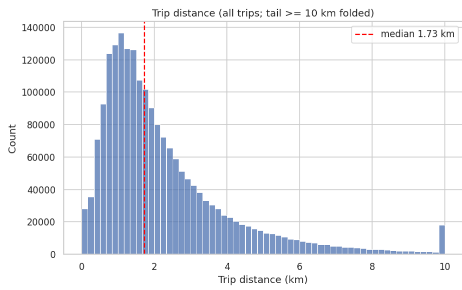

Figure 5: Distribution of trip straight-line distances.

Different Legacy Structure In 2020, Citi Bike updated its trip data schema, introducing column and formatting changes that differ from the 2013-2019 legacy format (detailed in our analysis notebook). Despite these structural discrepancies, the historical data remains still valuable, justifying the need for reformatting and refining them.

Missing Station Values A tiny fraction of trips contain missing values, confined entirely to station-related fields. The issue predominantly impacts end-of-trip attributes: in the representative month, roughly 4,800 trips (0.24%) lack an end station identifier, compared to just 0.03% for start stations. This gap likely stems from abandoned bikes or off-dock returns.

Trip Outliers Beyond missing values, a small number of trips represent clear outliers. In a representative month, approximately 315 trips (0.02%) last up to 25 hours, appearing in the final bin of the duration histogram (Figure 4). These likely reflect unreturned bikes rather than continuous rides. Additionally, around 27,000 trips (1.4%) start and end at the same station, resulting in a zero straight-line distance (Figure 5). Finally, a few trips record impossible distances, such as 11,000 km, due to corrupted station coordinates.

Biases This dataset carries certain inherent limitations. Because it relies exclusively on the Citi Bike system dataset, the findings may not accurately reflect the demographics of the entire population of cyclists in New York City. Additionally, trip distances are calculated using the Haversine formula between the start and end stations. This straight-line metric serves only as a geometric approximation and cannot account for the actual routes taken by riders, which may be longer due to street layouts, detours, or other factors.

2 Station Metadata Dataset (GBFS)

To enrich the static station data provided by Citi Bike, we incorporate their real-time feed, available on the Citi Bike System Data.

2.1 Dataset Structure

Unlike historical trip records, station metadata is served via a public JSON API adhering to MobilityData's GBFS standard. From this service, we query two key Citi Bike endpoints:

- `station_information.json`: Static properties like location and capacity, whose content changes only when the infrastructure is modified.
- `station_status.json`: Live availability and operational state.

2.1.1 Original Structure

The two endpoints share the `station_id` field, on which they are merged to produce a unified station record comprising:

- **Station ID**: Internal GBFS identifier, a UUID for most stations and a numeric string for the rest.
- **Short Name**: Public station identifier used for joins with Citi Bike trip data.

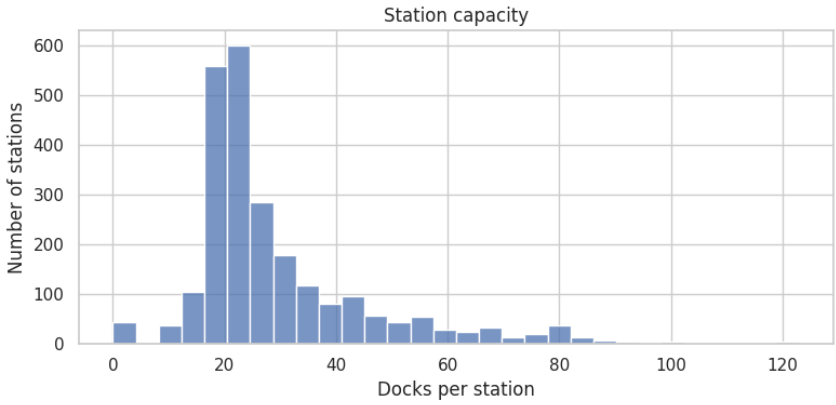

Figure 6: Distribution of station capacity, measured as the number of docks per station.

- **Name**: Station name.
- **Latitude** and **Longitude**: Station coordinates.
- **Capacity**: Total number of docks installed at the station.
- **Bikes Available**: Currently rentable bikes.
- **Vehicle Types Available**: Per-type counts of the bikes currently at the station.
- **Docks Available**: Currently available docks.
- **Operational Flags**: **is_installed**, **is_renting**, and **is_returning**.

2.1.2 Extracted Features

Using the three operational flags (`is_installed`, `is_renting`, and `is_returning`), we derive a boolean `active` indicator to filter out unavailable stations from live views. Evaluating all three status flags is necessary because physical installation does not imply service: stations can remain installed while suspending both rentals and returns. Relying solely on `is_installed` would therefore misclassify inactive stations as available.

2.2 Analysis and Insights

This section examines the structural and operational characteristics of the Citi Bike network, providing a foundational snapshot of its system infrastructure. Because the feed is a dynamic snapshot, this analysis represents a single point in time rather than a strictly reproducible state.

Capacity Distribution Figure 6 highlights the contrast between standard neighborhood stations and high-capacity transit hubs. The distribution peaks sharply between 18 and 28 docks, representing the typical capacity for residential and local commercial stations. Conversely, a extended right tail stretches beyond 50 docks (reaching up to 80 docks at major hubs) to accommodate heavy commuter turnover and intermodal connections. Notably, the count at 0 docks does not indicate corrupted data; rather, it represents 42 inactive stations currently present in the feed (as detailed in our analysis notebook).

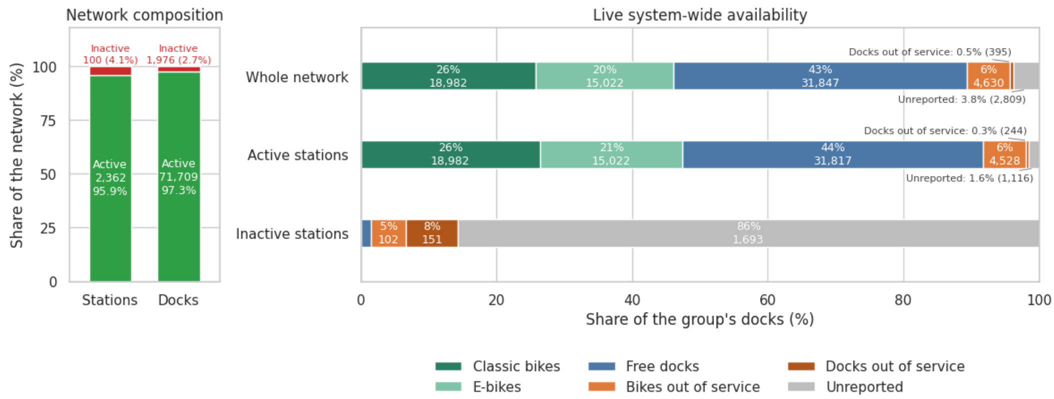

Figure 7: System-wide capacity breakdown across all, active, and inactive stations. The left panel gives the size of each group, in stations and in docks. In the right panel, total capacity is fully decomposed into six mutually exclusive states: rentable classic bikes, rentable e-bikes, free docks, out-of-service bikes, out-of-service docks, and unreported capacity.

Live Availability Figure 7 evaluates real-time operational capacity across the network. Unreported capacity represents the discrepancy when a station's static capacity exceeds the sum of its live bike and dock counts. It's instantly noticeable that a significant portion of the system's inconsistencies (out-of-service or unreported) is concentrated in the inactive stations, which represents only 4.1% of the total station count. Exposing supply-demand imbalances, bottlenecks, and maintenance backlogs would allow operators to quickly target areas for improvement.

2.3 Data Quality

To maintain data integrity and prevent visual distortion, our initial validation focused on those core quality issues. Ultimately, as confirmed in our analysis notebook, the live feed contains no missing station values.

Inactive Stations Stations undergoing installation, repair, or decommission are grouped under the `inactive` category, representing 100 out of 2,462 stations (4.1%) and 1,976 docks (2.7% of total capacity). As illustrated in Figure 7, these stations declare capacity while broadcasting almost no live counts, and the few bikes assigned to them are nearly all out of service. Retaining them would distort overall capacity metrics and render ghost (false empty) stations without contributing usable availability.

Unreported Capacity Unreported capacity (Figure 7) arises from inconsistencies between the two endpoints, which are not guaranteed to be synchronized in the updates. In the analyzed snapshot, the notebook records 2,809 unreported docks: 1,693 at inactive stations and 1,116 at active stations. It's possible to observe that this issue is particularly evident in inactive stations, but it also affects a small number of active stations, which may report fewer bikes and docks than their static capacity.

Identifier Consistency GBFS identifies stations in two ways: an internal `station_id` (either a UUID or a numeric string) and a public `short_name`. Trip history files use only the latter: 93.8% of station references in historical trips match a `short_name`, whereas none match a `station_id`.

Live-Only Metadata Citi Bike provides station metadata only as a live feed, with no historical archives. Consequently, removed stations retain their past trip records but lose all feed attributes, while newly added stations lack historical data, making them appear prematurely inactive. This bias only affects feed-specific attributes like capacity and operational status, as each trip record independently contains its own station name and coordinates.

3 Weather Dataset

To enrich the analysis with meteorological data, we incorporate hourly NYC weather data retrieved from the Open-Meteo archive API.

3.1 Dataset Structure

Weather data are queried via the Open-Meteo JSON API by specifying coordinates and date range. The API provides hourly data, updated with a few days' delay through an reanalysis model, offering a refined estimate of past conditions rather than keeping a live record of predicted weather.

3.1.1 Original Structure

The dataset includes several attributes. We focused on the following fields, as they are the one that most directly affect ridership:

- **Timestamp**: Hourly observation time.
- **Temperature**: Air temperature in °C.
- **Precipitation**: Hourly precipitation in millimetres.
- **Weather Code**: WMO weather condition code.
- **Wind Speed**: Wind speed in km/h.

3.2 Analysis and Insights

Weather conditions serve as a key driver of daily bike usage. Rather than examining weather in isolation, the following section focuses directly on its relationship with overall ridership patterns.

Ridership vs. Weather Figure 8 shows a clear contrast between how precipitation and temperature influence mobility patterns. Precipitation exhibits a strong inverse relationship with usage, where the month's peak precipitation directly coincides with its lowest ridership volume, yielding a Pearson correlation of $r = -0.55$. Conversely, daily temperature demonstrates a solid positive correlation ($r = +0.63$), with trip volumes closely tracking temperature rises. Quantified in our accompanying notebook, these relationships confirm that integrating weather data provides essential context for interpreting daily demand fluctuations and supporting operational planning.

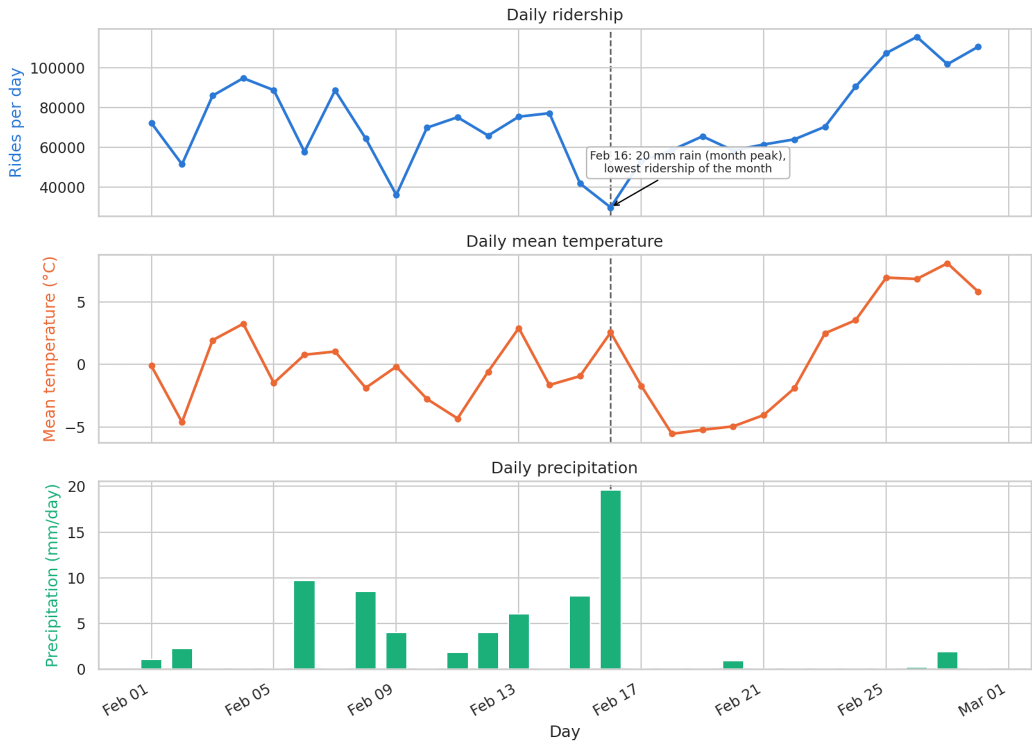

Figure 8: Daily ridership, mean temperature, and total precipitation over the reference month, plotted on aligned date axes with individual scales. The dashed guide highlights 16 February, the month's wettest day and also its least ridden.

3.3 Data Quality

While the dataset is complete with no missing values across the timeframe, these inherent limitations are still important to note.

Spatial Coverage Weather data is retrieved for a single coordinate centered on Central Park and applied across the entire system. Consequently, localized weather events, such as rainfall isolated to Central Park, are generalized citywide, which can occasionally misrepresent regional ridership patterns. However, while this single-point approach misses local microclimates, introducing higher spatial granularity would add unnecessary complexity for a city-scale analysis.

Timestamp Inconsistencies Aligning weather records with Citi Bike trip timestamps using local time introduces a one-hour shift during Daylight Saving Time (DST) transitions. This temporal mismatch temporarily desynchronizes weather observations from ride logs, introducing localized noise into the weather-ridership correlation.

4 Bike Lanes Dataset

To enrich our station and trip insights, we incorporate New York City's bike-route network, retrieved from the NYC OpenData portal.

4.1 Dataset Structure

Distributed as a single CSV file, the dataset provides a current snapshot of the bike network alongside a historical record, with the NYC Department of Transportation updating the file on an annual basis.

4.1.1 Original Structure

Each record represents a single segment of a bike facility, with the following fields:

- **Geometry**: The set of lines tracing the bike route segment, defined by their coordinates.
- **Street**: The street the facility runs along.
- **From Street** and **To Street**: The cross streets that define the segment's endpoints.
- **Borough**: Which of the five boroughs the segment lies in.
- **Facility Class**: The type of facility across four classes:
  - **Protected**: Separated from motor traffic by physical barriers (curbs, bollards, greenways).
  - **Conventional**: Marked by painted lanes only, without physical barriers.
  - **Shared / Signed**: Standard travel lanes with signage or sharrows indicating bicycle traffic.
  - **Link**: Non-bikeable connectors (stairs, walkways) bridging gaps in the network.
- **Status**: Whether the facility is still in service or has been retired/modified.
- **Installation Date** and **Retirement Date**: When the facility opened and, for facilities no longer in service, when it was removed or superseded.
- **Route ID**: The named route the segment forms part of.
- **Segment ID**: Which piece of street the record covers. The numbering comes from LION (*Linear Integrated Ordered Network*), the city's official street map, so every facility is pinned to a known piece of road.

Although finer classifications are recorded for each facility, they are excluded during ingestion. Because these sub-types aggregate naturally into four primary classes, omitting them preserves structural simplicity without affecting anything the visualization shows.

4.2 Analysis and Insights

The analysis provides a spatial and temporal perspective on the bike lane network.

Spatial Overview Figure 9 shows that the network covers almost the entire city, matching the distribution of demand trip areas shown in Figure 2. Comparing the two maps, we see that the densest and most popular areas are also the ones with the highest concentration of protected facilities, while shared and conventional lanes dominate in lower-demand areas. This alignment confirms the facility data's reliability, laying a solid foundation for integration with complementary datasets.

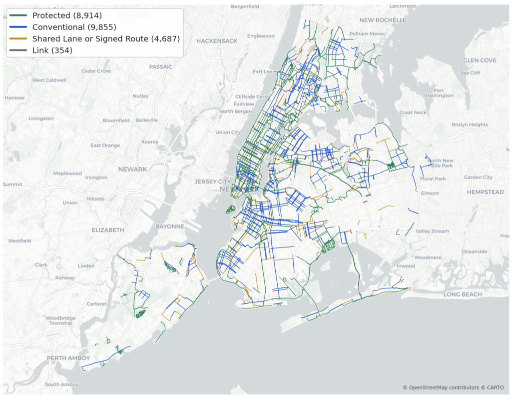

Figure 9: The bike-route network in service, colouring segments by facility class.

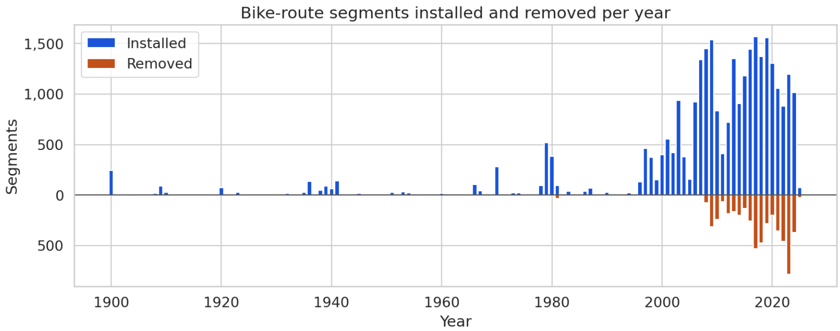

Figure 10: Segments installed and removed per year.

Temporal Overview Figure 10 illustrates the step-by-step expansion of the cycling network over time. Tracking annual installations and retirements reveals a sharp surge in infrastructure over the past decade, capturing a pivotal era in the city's urban transformation. Documenting both spatial and temporal dynamics provides key context for how past planning decisions continue to shape modern mobility.

4.3 Data Quality

With the exception of the retirement date (which is intentionally left blank for active facilities), all consumed attributes are fully populated, as verified in our analysis notebook.

Sentinel Installation Dates Although the installation date field is never null, `01/01/1900` acts as a floor for dates previous to the 1900s (Figure 10). It appears in 244 segments (0.8%), 196 of which remain active, concentrated on long-standing park and parkway paths such as Ocean Parkway and Eastern Parkway whose routes predate the city's record-keeping.

Facility Classification Bias Facility class reflects the highest class present along a segment, as reported in the documentation. For instance, a street with a protected lane on one side and a shared lane on the other is classified entirely as protected. This affects 720 segments (2.5%): a minor share that barely alters the map's overall coloring, but implies that segment classes represent the best available facility rather than conditions on both sides.